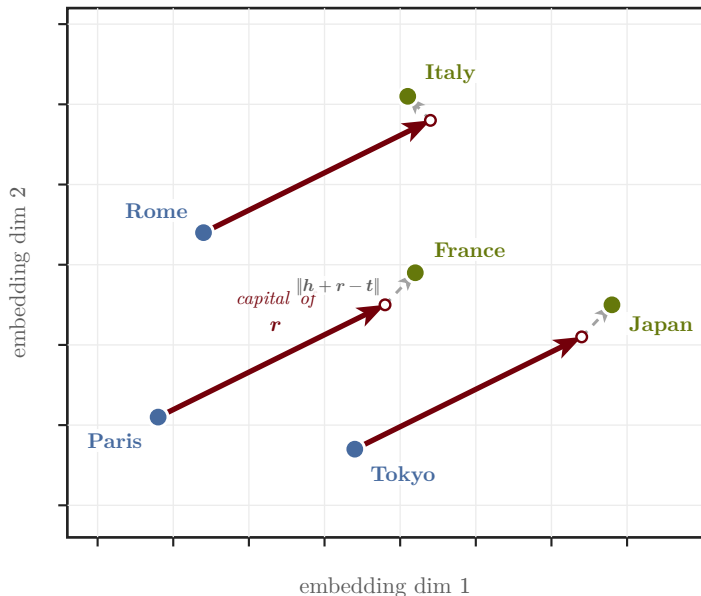


Knowledge-graph embedding — TransE

$$\mathbf{h} + \mathbf{r} \approx \mathbf{t}$$

one relation = one translation vector, shared across all its triples



Triples (h, r, t)

head h	relation r	tail t
Paris	<i>capital of</i>	France
Rome	<i>capital of</i>	Italy
Tokyo	<i>capital of</i>	Japan

Scoring a triple

$$f(h, r, t) = -\|\mathbf{h} + \mathbf{r} - \mathbf{t}\|_{1/2}$$

high score (~ 0): true triple

low score: corrupted triple (h, r, t')

Margin loss

$$\mathcal{L} = \sum [\gamma + d(\mathbf{h}\{+\}\mathbf{r}, \mathbf{t}) - d(\mathbf{h}\{+\}\mathbf{r}, \mathbf{t}')]_+$$

push true tail closer than a corrupted tail t' by margin γ

 relation translation $\mathbf{r}$

 residual $\mathbf{h}\{+\}\mathbf{r}\{-\}\mathbf{t}$

 head entity (capital)

 tail entity (country)